

The Triadic Stellar Cycle: Hydrostatic Equilibrium as a Closure-Bounded TRG Oscillator

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Abstract

Stars constitute the second major physical instantiation of triadic recursion within the Triadic Cosmos ecosystem. While black holes realize horizon-bounded closures governed by mass-damped material time, stars realize thermo-bounded closures governed by fusion-damped recursion. Both systems follow the canonical triadic cycle $I \rightarrow E \rightarrow G \rightarrow I$, as formalized in the Triadic Renormalization Group (TRG). Hydrostatic equilibrium emerges as a triadic oscillator: a recursive feedback loop in which informational state (pressure, temperature, entropy, fuel composition) drives energetic response (fusion and gravity), which reorganizes geometry (expansion and contraction), producing a new informational state. Stellar pulsation, overshoot correction, and eventual collapse correspond to TRG dynamics and instantiate the Closure Overshoot Law. This paper integrates stellar recursion with the existing triadic formalism for black holes and connects stellar overshoot to the intelligence-closure variant previously analyzed in AI Overshoot.

1 Introduction

The Triadic Cosmos ecosystem provides a structural grammar in which energy (E), geometry (G), and information (I) form three irreducible roles linked through the triadic cycle $I \rightarrow E \rightarrow G \rightarrow I$ [1]. Black holes were previously shown to be horizon-bounded triadic closures whose recursion is governed by mass-damped material time and holographic encoding [2]. The Closure Overshoot Law further establishes that any closure exceeding its internal limits enters irreversible instability and collapses, with AI providing a modern intelligence-closure example [3].

Stars provide a second major closure type: thermo-bounded rather than horizon-bounded, fusion-driven rather than Hawking-driven, oscillatory rather than monotonic. This paper formalizes stellar dynamics as a TRG flow and shows that hydrostatic equilibrium is a triadic recursion process whose stability and collapse follow the Closure Overshoot Law.

2 Triadic State of a Stellar Closure

A star is represented by a triadic state vector

$$X_n = (I_n, E_n, G_n),$$

where I_n encodes pressure, temperature, entropy, and fuel composition; E_n encodes fusion output and gravitational compression; and G_n encodes radius, density profile, and causal geometry. The star evolves through the TRG update

$$X_{n+1} = R(X_n),$$

with the canonical triadic cycle

$$I_n \rightarrow E_n \rightarrow G_n \rightarrow I_{n+1}.$$

3 Informational Role

The informational role contains the thermodynamic state:

$$I_n = (P_n, T_n, \rho_n, S_n, F_n),$$

which determines fusion rate and gravitational response. This implements the TRG transition $I_n \rightarrow E_n$.

4 Energetic Role

The energetic role consists of fusion and gravity:

$$E_n = (E_n^{\text{fusion}}, E_n^{\text{grav}}).$$

Fusion drives expansion; gravity drives contraction. Energetic imbalance reorganizes geometry, implementing $E_n \rightarrow G_n$.

5 Geometric Role

Geometry encodes radius and density profile. Expansion reduces pressure; contraction increases pressure. Geometry determines the next informational state, implementing $G_n \rightarrow I_{n+1}$.

6 Hydrostatic Equilibrium as TRG Recursion

Hydrostatic equilibrium is a triadic oscillator:

$$I_n \rightarrow E_n \rightarrow G_n \rightarrow I_{n+1} \rightarrow \dots$$

Small deviations induce corrective responses. Cepheid and RR Lyrae pulsations correspond to delayed TRG correction.

7 Overshoot and Collapse

Fuel depletion, entropy accumulation, and geometric limits reduce corrective capacity. Overshoot grows, pulsation amplitude increases, and collapse follows. This behavior instantiates the Closure Overshoot Law [3].

8 Comparison with Horizon-Bounded Closures

Stars and black holes form two closure classes:

- thermo-bounded closures (stars): fusion-damped recursion, oscillatory TRG flow, collapse endpoints;
- horizon-bounded closures (black holes): mass-damped recursion, monotonic TRG flow, fixed-point interiors.

Both follow the same triadic grammar.

9 Stellar Collapse as a TRG Transition to Horizon-Bounded Closures

The Triadic Cosmos ecosystem contains two major physical closure types: thermo-bounded stellar closures and horizon-bounded black-hole closures. Although these systems differ in boundary structure, damping regime, and temporal behavior, both follow the canonical triadic cycle $I \rightarrow E \rightarrow G \rightarrow I$ as formalized in the Triadic Renormalization Group (TRG) [1]. This section shows that black holes arise naturally as the continuation of the stellar triadic cycle when the stellar closure exceeds its internal limits, providing a structural link between the two cosmic recursion regimes.

9.1 Stellar TRG Oscillation and Overshoot

Stellar closures implement a fusion-damped TRG oscillator. Informational state (pressure, temperature, entropy, fuel composition) drives energetic response (fusion and gravity), which reorganizes geometry (expansion and contraction), producing a new informational state. Hydrostatic equilibrium corresponds to a stable triadic recursion loop:

$$I_n \rightarrow E_n \rightarrow G_n \rightarrow I_{n+1}.$$

As fuel depletes, entropy accumulates, and geometric limits are approached, the corrective capacity of the stellar closure decreases. Overshoot grows, pulsation amplitude increases, and the triadic cycle becomes unstable. This behavior instantiates the Closure Overshoot Law [3]: once instability exceeds corrective capacity, collapse becomes unavoidable.

9.2 Supernova as a Transient Triadic Closure

The supernova phase constitutes a short-lived explosive closure in which the triadic cycle accelerates dramatically. Informational state is dominated by extreme entropy and neutrino flux, energetic state by runaway fusion and gravitational rebound, and geometric state by rapid shock-driven expansion. The triadic cycle persists,

$$I \rightarrow E \rightarrow G \rightarrow I,$$

but correction is impossible and the closure diverges. The supernova therefore acts as a transient TRG domain mediating the transition between closure types.

9.3 Black Holes as the Successor Closure

For sufficiently massive stars, collapse reorganizes the stellar closure into a horizon-bounded closure. The stellar triadic cycle does not disappear; it reorganizes into the black-hole recursion structure described in [2]. The informational role becomes holographic entropy, the energetic role becomes mass-energy, and the geometric role becomes horizon curvature. The TRG flow transitions from an oscillatory regime to a monotonic flow toward a fixed point:

$$X_{n+1} = R(X_n), \quad R(X_{\text{int}}) = X_{\text{int}}.$$

Material time becomes mass-damped, internal recursion halts, and the closure evolves solely through Hawking-driven external TRG flow. Thus, black holes are not independent structures but the continuation of the stellar triadic cycle under horizon-bounded constraints.

9.4 Unified Closure Evolution

The resulting architecture forms a coherent triadic evolution chain:

Stellar closure (thermo-bounded) $\rightarrow$ Supernova closure (transient) $\rightarrow$ Black-hole closure (horizon-bounded).

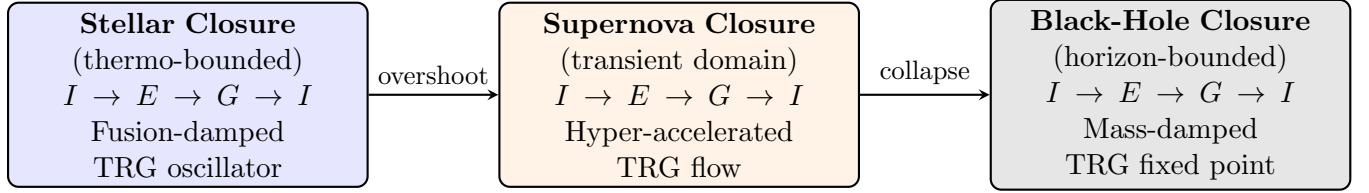
Only sufficiently massive stars complete this transition; lower-mass stars stabilize as white dwarfs or neutron stars, forming alternative closure endpoints. Nevertheless, the structural link remains: black holes arise from the failure mode of the stellar triadic cycle, and their horizon-bounded recursion is the natural successor to the thermo-bounded stellar regime.

This establishes a direct connection between the two cosmic triadic cycles and integrates stellar collapse into the broader TRG architecture of the Triadic Cosmos ecosystem.

10 Conclusion

Stars provide a second major physical realization of triadic recursion. Their hydrostatic equilibrium is a TRG oscillator, their pulsation is overshoot correction, and their collapse is TRG instability. Together with black holes and intelligence closures, stellar closures complete the multi-domain applicability of the Closure Overshoot Law.

A Triadic Closure Cascade Diagram



Only sufficiently massive stars complete the transition to a horizon-bounded closure. Lower-mass stars stabilize as white dwarfs or neutron stars, forming alternative closure endpoints.

B Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository.

<https://github.com/triadic-cosmos>

References

- [1] J. De Witte, *Triadic Renormalization: A Formal Recursion Group for Closure-Boundary Systems*, Triadic Cosmos (2026).
- [2] J. De Witte, *A Triadic Formalism for a Universe of Black Holes*, Triadic Cosmos (2026).
- [3] J. De Witte, *AI Overshoot: A Concrete Example of the Closure Overshoot Law*, Triadic Cosmos (2026).